Black holes have the universe's most inscrutable poker faces. From the outside, they appear just about as simple as you can get. The three distinguishing features of a black hole are its mass (which determines how big it is—the distance from its center to its event horizon, the enshrouding surface of no return), its electric charge, and how fast it's spinning. That's it. There are no more details to be gleaned from scrutinizing the visage that a black hole presents to the cosmos. Physicists sum this up with the saying "Black holes have no hair," meaning that they lack the kinds of detailed features that allow for individuality. When you've seen one black hole with a given mass, charge, and spin (though you've learned these indirectly, through their effect on surrounding gas and stars, since black holes are black), you've definitely seen them all.

Nevertheless, behind their stony countenances, black holes harbor the greatest reservoirs of mayhem the universe has ever known. Among all physical systems of a given size with *any* possible composition, black holes contain the highest possible entropy. Recall from Chapter 6 that one rough way to think about this comes directly from entropy's definition as a measure of the number of rearrangements of an object's internal constituents that have no effect on its appearance. When it comes to black holes, even though we can't say what their constituents actually are— since we don't know what happens when matter is crushed at the black hole's center—we can say confidently that rearranging these constituents will no more affect a black hole's mass, charge, or spin than rearranging the pages in *War and Peace* will affect the weight of the book. And since mass, charge, and spin fully determine the face that a black hole shows the external world, *all* such manipulations go unnoticed and we can say a black hole has maximal entropy.

Even so, you might suggest one-upping the entropy of a black hole in the following simple way. Build a hollow sphere of the same size as a given black hole and fill it with gas (hydrogen, helium, carbon dioxide, whatever) that you allow to spread through its interior. The more gas you pump in, the greater the entropy, since more constituents means more possible rearrangements. You might guess, then, that if you keep on pumping and pumping, the entropy of the gas will steadily rise and so will eventually exceed that of the given black hole. It's a clever strategy,